Centre for Distance and Online Education
University of Calicut

BA HISTORY HONOURS

HS1CJ01
History of Keralam Upto the 12th Century

E-Content

Module II
Keralam Upto the Iron Age

Chapter 1
Prehistoric Settlements : Palaeolithic Mesolithic and Neolithic

PALAEOLITHIC CULTURES

The Palaeolithic period, meaning the Old Stone Age, represents the earliest and longest phase of human cultural history. Spanning from the first manufacture of stone tools to the end of the last Ice Age, the Palaeolithic covers nearly 99 per cent of human existence. It marks the formative stage during which early humans learned to make tools, adapt to diverse environments, and develop basic survival strategies such as hunting, gathering, and scavenging. Since this period long predates the invention of writing, our understanding of Palaeolithic cultures across the world is based almost entirely on stone tools, fossil remains, and their geological and environmental contexts.

Globally, Palaeolithic cultures developed during the Pleistocene epoch, a time characterised by dramatic climatic fluctuations, including repeated glacial and interglacial phases. These climatic changes reshaped landscapes, altered vegetation and animal populations, and directly influenced patterns of human movement and settlement. As a result, Palaeolithic cultures across different continents were not uniform but showed regional diversity shaped by environment, raw material availability, and ecological constraints. Despite this diversity, certain broad technological and behavioural trends allow archaeologists to study the Palaeolithic as a coherent global phenomenon.

Africa occupies a central place in the global Palaeolithic narrative, as it is widely recognised as the earliest centre of stone-tool manufacture. Some of the oldest known stone tools have been discovered in eastern and southern Africa, particularly at sites such as Olduvai Gorge, where early stone industries demonstrate the beginnings of human technological behaviour. From Africa, early human populations gradually dispersed into other parts of the world, carrying with them stone-tool traditions that were adapted and modified in new environments.

As Palaeolithic cultures spread into Europe and Asia, they encountered vastly different ecological zones, including temperate forests, open grasslands, deserts, and cold Ice Age environments. In Europe, Palaeolithic cultures are often associated with cave sites and show well-defined technological sequences, particularly during the later phases of the period. In contrast, large parts of Asia, including South and East Asia, display extensive open-air sites, river-valley occupations, and long-lasting stone-tool traditions that do not always follow European models. This has led scholars to emphasise that the global Palaeolithic cannot be understood through a single evolutionary pathway but must be seen as a mosaic of regional adaptations.

Across the world, the Palaeolithic period is broadly divided into Lower, Middle, and Upper Palaeolithic stages, reflecting gradual changes in stone-tool technology and human behaviour. The Lower Palaeolithic is generally characterised by large core tools such as choppers and hand-axes, the Middle Palaeolithic by flake-based industries with improved control and efficiency, and the Upper Palaeolithic by blade tools and greater technological sophistication. However, these divisions are analytical frameworks rather than rigid stages, and their chronological boundaries vary from region to region.

A key feature of Palaeolithic cultures in the global context is the close relationship between humans and their environment. Palaeolithic groups were highly mobile and depended directly on natural resources for survival. Their tools were primarily designed for functional purposes such as cutting, scraping, and hunting rather than for symbolic or decorative use. Over time, improvements in tool-making techniques

reflect growing cognitive abilities, planning skills, and social cooperation, even though evidence for art and symbolic behaviour appears unevenly across different regions.

In recent decades, global Palaeolithic studies have moved away from Eurocentric models that once treated European sequences as universal standards. Archaeological research in Africa, Asia, and Australia has demonstrated that early humans developed multiple technological solutions to similar survival challenges. This shift has been particularly important for understanding regions such as South Asia, where long-lasting pebble-tool traditions and unique technological patterns challenge traditional evolutionary assumptions.

Palaeolithic Cultures in India

The Palaeolithic period marks the earliest phase of human history in India, representing a very long span of time during which humans lived entirely by hunting, gathering, and scavenging. This period belongs to the Pleistocene epoch, an age characterised by repeated climatic fluctuations, including glacial and interglacial phases. Since there are no written records and almost no human skeletal remains, the study of Palaeolithic cultures in India depends primarily on stone tools, their typology, manufacturing techniques, and their association with geological deposits such as river terraces and gravel beds. Unlike Europe or Africa, where cultural sequences often show sharp transitions, the Indian evidence suggests slow and continuous development strongly influenced by geography, climate, and the availability of raw materials. The following sections examine the Lower, Middle, and Upper Palaeolithic stages.

Lower Palaeolithic Cultures in India

The Lower Palaeolithic represents the earliest and longest phase of the Palaeolithic in India. It is characterised by the use of large, heavy stone tools made primarily on cores and pebbles. During this phase, early humans lived in close association with river valleys, open grasslands, and forest margins, which provided water, animals, and raw materials for tool-making.

A striking feature of the Indian Lower Palaeolithic is the regional diversity of tool traditions. In north-western India and parts of present-day Pakistan, the dominant tradition is the pebble-tool or Soan culture, while in peninsular India the Acheulian hand-axe tradition is more prominent. This duality is a distinctive characteristic of the Indian Lower Palaeolithic.

The Soan culture is primarily associated with river systems in the north-west. Tools were made from rounded pebbles collected from riverbeds. These tools include choppers, chopping tools, flakes, and cores. They are generally crude in appearance, with limited secondary working, suggesting that tool-making was aimed at immediate functional needs rather than aesthetic form. The long continuity of this pebble-tool tradition indicates that it was well adapted to local environmental conditions.

In contrast, the Acheulian tradition, better represented in peninsular India, is characterised by hand-axes, cleavers, and picks. These tools show greater planning, symmetry, and control in manufacture. Acheulian tools were usually made from large stone blocks of quartzite or basalt and are associated with open landscapes where large animals were available for hunting or scavenging. Unlike Africa and Europe, Acheulian tools in India are not uniformly distributed and often occur alongside or separate from pebble-

tool industries. This has led scholars to argue that India followed an independent and regionally varied Lower Palaeolithic trajectory.

Middle Palaeolithic Cultures in India

The Middle Palaeolithic represents a technological and behavioural transition from the earlier phase. During this period, there is a clear shift from large core tools to smaller, flake-based implements. The emphasis moved from shaping the core to producing flakes of desired size and form, indicating improved control over tool-making techniques.

Middle Palaeolithic tools include scrapers, points, borers, and denticulates. These tools are lighter and more specialised than those of the Lower Palaeolithic, suggesting changes in subsistence strategies and daily activities. Hunting techniques likely became more efficient, and tools were increasingly used for processing animal skins, cutting meat, and working plant materials.

This phase is widely distributed across India, including river valleys, rock shelters, and open-air sites. The Middle Palaeolithic is often associated with early forms of Homo sapiens and reflects increasing adaptability to diverse ecological zones. Despite these changes, the transition from the Lower Palaeolithic was gradual, with earlier traditions continuing in modified forms.

Upper Palaeolithic Cultures in India

The Upper Palaeolithic marks the final stage of the Palaeolithic period and shows significant technological advancement. The most important feature of this phase is the development of blade technology, in which long, narrow flakes were deliberately produced from prepared cores. These blades were then modified into a variety of tools such as burins, end scrapers, and points.

Upper Palaeolithic tools are smaller, lighter, and more efficient, reflecting a higher degree of planning and skill. There is also limited evidence for the use of materials other than stone, such as bone or antler, although stone tools remain dominant. Hunting strategies appear to have become more organised, possibly involving group coordination and seasonal movement.

Rock shelters and cave sites from this period sometimes show evidence of repeated occupation, suggesting more structured use of space. Although artistic and symbolic expression is not as prominent as in European Upper Palaeolithic contexts, the Indian evidence nevertheless points to increasing cognitive and cultural complexity.

Environment, Subsistence, and Way of Life

Throughout the Palaeolithic period, Indian hunter-gatherers lived in close relationship with their environment. River valleys played a crucial role by providing water, animals, and stone for tools. Animal remains found at Palaeolithic sites include large herbivores such as elephant, bison, horse, and deer, indicating that hunting and scavenging formed the basis of subsistence.

Climatic fluctuations during the Pleistocene repeatedly altered landscapes, vegetation, and animal populations. Palaeolithic cultures in India reflect continuous adaptation to these changes rather than

abrupt cultural breaks. Mobility was a key feature of life, with groups moving seasonally in search of food and resources.

Significance of Palaeolithic Cultures in India

The Palaeolithic cultures of India are significant because they represent the earliest evidence of human presence and activity in the subcontinent. They demonstrate how early humans adapted to a wide range of ecological conditions, from river valleys and plains to rocky plateaus and forested regions. The long continuity of stone-tool traditions highlights the importance of local environments in shaping cultural development.

Scholars such as H. D. Sankalia have emphasised that Indian Paleolithic cultures should not be seen as mere reflections of European or African sequences. Instead, they represent an independent and regionally distinctive prehistoric tradition, marked by gradual change, technological flexibility, and close interaction with nature.

Major Paleolithic Tools and Their Regional Distribution in India

The Paleolithic cultures of India are best understood through the regional distribution of stone tools, as tool types vary considerably across different parts of the subcontinent. This variation is closely related to environmental conditions, river systems, availability of raw materials, and subsistence patterns. Instead of a single uniform technological tradition, the Indian Paleolithic shows multiple regional adaptations, each reflected in the kinds of tools produced and used by early human groups.

In north-western India, particularly in the Soan (Sohan) Valley and adjoining regions of Punjab, the dominant Paleolithic tools belong to the pebble-tool tradition. Here, early humans primarily used naturally available river pebbles to manufacture tools such as choppers, chopping tools, flakes, and simple cores. These tools are generally crude in appearance, with limited secondary working, and were designed mainly for basic activities like cutting meat, breaking bones, and processing plant materials. The rarity of hand-axes in this region suggests that pebble tools were sufficiently effective for survival in a river-dominated landscape. The long continuity of this tradition indicates a stable adaptation to local environmental conditions rather than technological backwardness.

In contrast, peninsular India, especially the Deccan Plateau and river valleys such as the Godavari, Krishna, and Tapi, is characterised by the widespread presence of Acheulian hand-axe tools. The most distinctive tools from this region are hand-axes, cleavers, and picks, which display a higher degree of planning and symmetry compared to pebble tools. These implements were generally made from large quartzite or basalt blocks available in abundance in the region. The Acheulian tools suggest a technological adaptation suited to open grasslands, where hunting or scavenging of large animals was possible. Unlike the north-western region, hand-axes in peninsular India appear as a dominant tradition rather than a marginal one.

The Narmada Valley in central India occupies a particularly important place in Paleolithic studies because it shows evidence of both early and later Paleolithic technologies. Stone tools

recovered from this region include Acheulian hand-axes and cleavers as well as Middle Paleolithic flake tools such as scrapers and points. This diversity reflects a long sequence of occupation and gradual technological transition. The Narmada Valley, with its extensive river terraces and rich faunal remains, appears to have functioned as a major corridor for human movement and cultural development in prehistoric India.

In Rajasthan, especially in the Didwana region and areas associated with ancient lake beds and desert margins, Paleolithic tools are mainly flake-based and belong largely to the Middle and Upper Paleolithic phases. The most common tools found here are scrapers, points, borers, and blades. These tools are smaller and lighter than earlier core tools, indicating increasing efficiency and adaptability. The presence of Paleolithic tools in what is now a desert region shows that the climate in the past was more favourable and that early humans were capable of adjusting their technology to semi-arid environments.

The eastern regions of India, including the Chotanagpur Plateau and parts of Odisha and Jharkhand, present a mixed Paleolithic assemblage. Both pebble tools and Acheulian hand-axes are found, along with flake tools from later phases. The tools in this region were typically made from locally available raw materials such as quartz and quartzite. The technological diversity seen here suggests that eastern India acted as a zone of interaction where different tool traditions overlapped rather than one where a single dominant industry developed.

In southern India, particularly in Tamil Nadu, Karnataka, and Andhra Pradesh, the Paleolithic record is extensive and shows a clear technological sequence from the Lower to the Upper Paleolithic. Early levels are dominated by Acheulian hand-axes and cleavers, while later levels show Middle Paleolithic scrapers and points, followed by Upper Paleolithic blade tools. The long continuity of occupation in this region indicates that southern India was an important centre of early human settlement, supported by favourable ecological conditions and abundant raw materials.

Rock shelters and cave regions, such as those in central and southern India, yield mainly Middle and Upper Paleolithic tools, including scrapers, points, blades, and burins. These sites are significant not only for the tools found but also for what they reveal about settlement patterns. The repeated use of rock shelters suggests seasonal movement and more organised use of space, marking an important development in human behaviour.

Overall, the regional distribution of Paleolithic tools in India demonstrates that early humans developed flexible and adaptive technologies suited to specific environments rather than following a single technological path. Pebble tools, hand-axes, flake tools, and blade industries often co-existed in different regions and periods, reflecting local needs and resource availability. This diversity underlines the importance of understanding the Indian Paleolithic as an independent and regionally varied cultural tradition, shaped by environmental conditions and long-term human adaptation.

Palaeolithic Cultures in south India

Studies in the prehistory of India started with the discovery of a hand axe by Robert Bruce Foote in 1863 in the vicinity of Pallavaram, near Chennai (Madras). Since then, in addition to his own

surveys in different parts of the Madras Presidency, the explorations of many other archaeologists during the past one and a half centuries have filled up the prehistoric map of India and have more or less linked it to developments in world prehistory. Also, the prehistory of different parts of India has been integrated well on the basis of geo-archaeological studies of major river systems coupled with other archaeological methods such as the study of tool manufacturing techniques. In field studies, certain scholars have emphasized the understanding of hominid behaviour in various regional settings by pro-longed observation of certain least-disturbed fossilized archaeological sites. Some sites have been subjected to systematic excavation too, such as Attiram-pakkam in Tiruvallur district northwest of Chennai and in the Hunsgi valley in the middle reaches of the Krishna in Gulbarga district. The present understanding of the stone-tool-using cultures of India is that the Stone Age may be classified into two major divisions: Palaeolithic and Neolithic. The Palaeolithic in turn is subdivided into lower, middle, and upper Palaeolithic cultures, followed by the Mesolithic culture. In geological chronology, the Paleolithic cultures start from the middle Pleistocene period and go on to almost to the end of the Pleistocene period and even into the first half of the Holocene period.

The lower Paleolithic culture is characterized by Acheulian-type tools. During the early stages, the tools are crude and heavy and have zigzag cutting edges. In the later stages the tools are thinner and less heavy, fully flaked with little cortex, and have straight cutting edges. They are more standardized and, in addition to hand axes and cleavers, the late Acheulian tools also contain scrapers of various types made from cores and flakes. The tools were believed to have been used for various functions like hunting, butchering, digging of roots, and making wooden implements.

The associated faunal remains of this culture, so far known from the river valleys of the Narmada and Godavari, are elephant, horse, cattle, and wild boar indicating a palaeo-environment marked by moist deciduous woodlands and grasslands. Radiometric dating of the bones of faunal remains and volcanic ash puts the Acheulian phase's time-span berween 1.2 million years to one hundred and fifty thousand years (150,000 years). The discovery of parts of a fossilized human cranium on the banks of the Narmada dated to between five hundred and thousand years (500, 000 years) and two hundred and thirty six thousand years (236,000 years), in association with middle-to-late Acheulian industries, though isolated so far, has given rise to some interesting discussions related to the dispersal of hominids from Africa and the origin of modern humans (Homo sapiens).

Tools of the middle Palaeolithic period, which was christened 'Nevasan' by Sankalia, have been discovered in different parts of south India, showing human adaptation to varied environments. Some of the discoveries were made at occupation sites. The tools are smaller in size and mostly prepared from the detached flakes of pebble cores or big stones and include small-sized hand axes, scrapers of various types, borers, points, and choppers. Some important sites are Nevasa, Adamgarh Hill (Maharashtra), Sanganakallu (Karnataka), Renigunta, Kurnool, Nagarjunakonda (Andhra Pradesh), and Attirampakkam, Gudiyam, and Vadamadurai (Tamil Nadu). The tools are assigned to the late middle Pleistocene to upper Pleistocene times on geomorphologic grounds.

The upper Palaeolithic culture (40,000 to 8,000 BP) is found widely in several cave sites in the river valleys of the southern part of the Eastern Ghats and in other parts of peninsular India. The

distinctive feature of the stone tools of this culture is the production of blades from prismatic or fluted cores. Other tools comprise burins, awls, and a variety of flake and core tools, including various kinds of scrapers. The tools being very small must have been mounted on wooden or bone shafts using resins. Fine-grained quartzite, chert, jasper, chalcedony, and limestone were the raw materials used for making them. The bone-tool industry was an important trait of this culture, as is evident from the excavation of the Billa Surgam caves in the Kurnool area. Though in the Vindhyan cave sites much rock art is found associated with the upper Palaeolithic culture, we do not see any art remains in the south. The associated faunal remains from the Kurnool caves belong to the late Pleistocene age. The animals identified so far are nilgai, antelope, wild boar, tiger, leopard, sloth bear, and the jungle cat.

The upper Paleolithic culture was followed by the Mesolithic culture, sometimes inappropriately called the Microlithic culture due to a preference for Microlithic tools at this stage. Sites of the Mesolithic culture are found in varied physiographical zones-coastal areas, riverine eco-zones, plateaus, and woodlands. The teri sites of the southeastern coast of Tamil Nadu are well-known sites of this culture. Technologically, the Mesolithic culture is an extended development of the upper Palaeolithic tradition, producing blade and bladelet tools by pressure retouch. The common artefacts are backed points, blades, lunates, triangles, and trapezes. While quartz is a predominant raw material for making Microlithic tools in Kerala, in most other parts of south India, besides quart,. chert, jasper, chalcedony, agate, and crystal are also used. It has been observed that in some sites like the Kurnool caves, the raw material for making blades has been fired. Besides the microliths, sling balls, quarrzite hammer stones, and some heavy-bored spherical stones were also part of the tool equipment. From the paintings in Vindhyan rock shelters it may be inferred that the bow and arrow was a common hunting weapon during this period. From this and an assemblage of Microlithic tools, which could have been mostly compound tools made with hafted or strapped wooden or bone shafts been versatile hunter-gatherers in various environments. They also occupied themselves in fishing. Radiocarbon dates to the Mesolithic sites on the Vindhyan Hills suggest a time range from 8500 BCE to 2000 BCE. On typological comparison, this time range can be applied to south Indian sites as well. However, in certain Neolithic sites, and Iron Age sites too, Mesolithic tools have been found, suggesting the survival of hunter-gathering communities along with agro-pastoral communities of the Neolithic and Iron Ages.

Palaeolithic Cultures in Kerala

Earliest evidence of ancient human habitation in Kerala dates to the Upper Paleolithic Age (25000 to 30000 years). For convenience, the Stone Age in India is divided into the Lower Paleolithic and Upper Paleolithic periods, each distinguished by characteristic features. As elsewhere, is the prevalence of a specific kind of pebble-butted axe made out of large pebbles found in river beds. The stone axe has three of its sides, barring the handle, form rough curved edges by flaking. Such stone axes have been found only on the banks of river beds in North West India and the Deccan. In the Lower Paleolithic Age, people lived in natural rock shelters at high altitudes and on big game-hunting with the fleet-footed deer in the prairies of the rainforests. Lower Paleolithic antiquity goes back to five lakh years from the present, dated through ESR test on the stone tools discovered at the archaeological site of Athirappakkam near Chennai. Upper Paleolithic antiquities of south India are presumed to be 40,000 to 50,000 years old. Mesolithic sites (30000-25000 BP) with paintings are assigned to 25,000 to 38,000 years from the present.

A slightly late phase of the Upper Paleolithic Age is called Mesolithic Age and it falls between the Upper Paleolithic and late Stone Age (or Microlithic Age).

Paleolithic people could survive only in high altitude environments. Originally they co-existed with mammoths that lived in high altitude icy mountains and hunted this huge animal using stone-tipped spears on a face-to-face encounter. Sometimes they drove the animal into swamps or other landscape traps. Since all this involved lot of collective physical combat with the animal, archaeologists called it big game hunting. Recent studies have shown that they used to scavenge too. They are said to have scavenged the bone prairies from the animal waste left by big predators. There is no such high altitude environment of ice-capped mountains in the Western Ghats of Kerala and signs of human habitation going back to such a hoary antiquity. However, recent studies have shown the presence of the people of the Upper Palaeolithic Age in the high ranges of Kerala.

Stone Tools

Some Upper Paleolithic tools like choppers have been reported recently by P. Rajendran in his archaeological studies along the banks of the river Kanjirapula in Palakkad district. Similar studies on tools by Padmanabhan Thampi in the Koodakkad reserve forest of Marayur have shown the presence of habitation during the Mesolithic period. In May 1982 more pieces of evidence surfaced at a few places on the right bank of the river Tirthakkara and in the caves of the Chenthuruni hills that is surrounded by the tributaries of the river Achenkovil. Upper Paleolithic implements such as choppers flaked out of pebbles have also been discovered in Kerala. These are relatively small flake-tools with curved edges showing features of local distinction. Normally the core was preserved after use while the choppers were abandoned on river banks where the hunters process the hunt. These people might have lived on hunting the gaur, tahr and deer, probably by using the stone tools hafted to long sticks. Such tools were common during the Mesolithic phase of the Stone Age, which is well-known for rock-paintings.

Upper Paleolithic and Mesolithic antiquities of Kerala are not scientifically dated as yet. What we have is a relative dating based on comparative archaeology applied to the stone tools and paintings of the sites in Andhra, Karnataka and Tamil Nadu. Accordingly, archaeologists have presumed the Upper Paleolithic and Mesolithic tools and paintings to be 25,000 to 30,000 years old.

MICROLITHS IN KERALA

In Kerala what follows the 'chopper' culture is the Late Stone Age period noted for its small stone tools or microliths. It appears that the habitat of people during the Late Stone Age period was the red soil slopes and marshy plains where they subsisted on trapping smaller animals, birds and fish. Naturally, their tools were smaller. Microliths were reported from the quartzite tracts in the plains of Kerala, because these tools are made of small quartz chips flaked from rock dykes with the help of a core and shaped into tiny stone blades and points. These could be hafted to wooden sticks or bones for use as axes, knives and sickles. Sticks fitted with stone tips might have been used to hunt animals, the blades to scrap animal skin and the sickle to reap wild cereals. In Kerala, microliths made of quartzite flints alone have been found, whereas in other places in India and elsewhere microliths made of semiprecious stones such as carnelian, lapis lazuli and jasper have also been found. Kanhirapally in Kottayam district, Marayur in Idukki,

Keezhillam in Ernakulam and Kanjirapula and Anakkara in Palakkad are some Microlithic sites in Kerala.

As in the case of the Upper Paleolithic period, Microlithic tools of Kerala have not been scientifically dated because they have not been discovered in any precisely datable stratified context. Using comparative archaeological method, therefore, archaeologists have tentatively dated the microliths of Kerala as being at least 10,000 years old.

Overlap in Rock Art

The Mesolithic culture in Kerala seems to have continued till around 3000 BCE from where probably the Microlithic culture begins. As regards the question of successive Stone Age cultures, it is to be borne in mind that as a general rule a technologically superior one succeeds one that is at a lower level. However, this need not necessarily imply the complete disappearance of the inferior type with the inception of the superior one. They often co-existed, as is the case even today. Along with people of the Microlithic phase, the chopper culture existed, but in their own habitats. These sequences as well as the coexistence are attested by the multilayered rock-art.

All the major rock-paintings in Kerala are multilayered and obviously belong to disparate phases from the Mesolithic down to the Iron Age. It is quite possible that there are Microlithic superimpositions on Mesolithic rock-paintings and Neolithic and Iron Age superimpositions on the Microlithic. The Marayur hood rock-painting, for instance, is a typical example. Here, we see a big human figure with wavy lines, a graffiti very frequently met with at many places in south India and Sri Lanka on Iron Age pottery. However, this depiction is superimposed over certain other depictions, probably of three different cultures. There are several dolmens that represent huge rock-cut burial monuments, indicating the possible connection to the Iron Age. There about two hundred depictions of dolmens all along the surface of the Alampaṭṭi rock. There are red ochre paintings at various dolmen sites of south India. Similarly, the rock-art at sites such as Tovari, Eaṭakal and Ankode is layered, representing cultures probably from the Microlithic and Neolithic to the Iron Age, through depictions with indications of hunting, domestication and cattle-keeping.

Engravings at Tovari and Eaṭakal rock-shelter, located on the hill called Ampukuttimala in the Wayanad district, are with certain comparisons and contrasts. Tovari engravings are smaller and thinner incisions on a spherical stone with a roof-like structure. They depict implements such as stone arrow-heads and crow bar of triangular shape, set in such geometrical shapes as squares and circles. There is also a drawing seemingly that of a stylised bind. Since these incisions are too thin to be made with a stone axe, it is reasonable to presume that they were made with microliths and, therefore, slightly anterior to the Eaṭakal engravings. At the same time, the crow bar drawn along with a square divided into eight segments seems to depict fields of terraced farming.

Eatakal, literally meaning the gap between stone blocks, is a huge natural rock-shelter. Since these are weathered rock, they can easily be carved using a stone axe. Several pictures can be found engraved on the rock-walls on either side, in different styles enabled by deep and shallow incisions, most of which are of 1-2 cms wide and 2 cms deep with shallow edges on both ends.

There are human figures in postures of a ritual dance, and animals such as elephants, wild dogs and deer in the forest environment. Moreover, there are wheeled carts with the driver, various images, geometric forms and orthographic symbols. Neolithic hand-axes were reported to have been collected from the site by Fred Fawcett who discovered the rock-engravings in the cave in 1901 CE.

Ankode, locally known as Pandavampāra, is located at an elevation of 1,000 feet in the village of Perumkadavila of the Neyyattinkara taluk in Thiruvananthapuram district. It is with the entire surface easily amenable to deep engraving weathered rock comparable to the rock of Tovari and Eatakal Neolithic hand-axes were reported to have been recovered from wide, which could be executed using a sharp stone flake such as the site. The Ankode engravings are 1.9 cm deep and 1.5 cm wide. However, no comparison can be made between the figures at Ankode and those at Edakkal, as the Neolithic hand-axe and other tools reported from Ankode are too limited in both quantity and diversity.

Neolithic Culture in south India

The Neolithic culture is considered a very important stage in human progress as it marked the first stage of the introduction of agriculture and food crops through human effort. Within India, four distinct Neolithic culture zones, each having its own chronology and cultural traits, have been found. The Neolithic culture of south India, simply the southern Neolithic, is found over a wide area covering a major part of Karnataka and Andhra Pradesh (Belgaum, Bijapur, Gulbarga, Raichur, Bellary, Kurnool, Anantapur, Tumkur, Bengaluru [Bangalore], and Mysuru (Mysore) districts) and the northern districts (Dharmapuri, Krishnagiri, Salem, and North Arcot) of Tamil Nadu. In the central and southern parts of Tamil Nadu, and also in Kerala, no concrete evidence is so far available for the existence of a true Neolithic culture. A concentration of Neolithic sites is found in the middle and upper reaches of the Krishna-Tungabhadra river system. Robert Bruce Foote, who discovered numerous sites in the Karnataka part, inaugurated the serious study of this phase too. Its stratigraphical-chronological position was made Brahmagiri. Subsequently, a number of significant excavations took place at the Neolithic sites of Sanganakallu, Maski, Piklihal, Nagarjunakonda, and so on, and several intensive regional surveys were under-taken in the Andhra and Karnataka regions. These studies have thrown much new light on various aspects of this culture: settlement patterns, nature of dwellings, food grains, burial practices, ceramic and stone tool traditions, and also on the ash mound appurtenances of the sites. While the chronology of the different phases of this culture is more or less settled, the origins of Neolithic culture are not clear; the settlements here seem to appear suddenly about 3000 BCE while the other neolithic cultures in India appear to have some antecedents. The Neolithic culture lasted till about 1000 BCE, but some remnants well, and this is attested to by the occurrence of Neolithic hand axes (celts) in Tamil Nadu and Kerala in Iron Age contexts.

The people of the Neolithic culture were primarily cattle-keepers and to some extent cultivators of a range of millets and other crops. They lived close to small streams rather than major rivers. The early settlements were just stockaded cattle pens in open country. Later, the settlements were located on artificially made terraces on the granite hills, and consisted of round huts made of wattle and daub supported by wooden posts, and with earthen floors. The heavy stone equipment of the Neolithic culture was made of basalt and included numerous finely made axes with elongated triangular out-lines and cylindrical sections. Saddle querns, hammer stones, and sling

balls were some of the household equipment. Their tools and equipment also included fine, regular blades made of chert. Although copper is not used much in the early stages, its use is not insignificant in the later stages. Turntables were used to make ceramic wares, which included grey, buff, or brown burnished pottery, with a small quantity of red or black slipped ware, some with painted designs. Urn burials have been found within the habitations. Most of the burials known so far are those of children, with a few exceptions containing adult remains. Burial goods included stone axes, chert blades, and pots. The few available skeletons of adults show they were of tall and large-bodied people. Since the skeletal material is scarce, the conclusions drawn from earlier physical anthropological studies of the material with regard to racial affinities are difficult to generalize and the recent critical review of them show that the basic premises and methods of those studies are ill-founded. In the past few decades bio- and dental-anthropological studies are being carried out to understand issues demography and diseases rather than of racial features and affinities.

The economy of this culture, as mentioned in the foregoing, was centred on cattle-keeping and cattle pens are a prominent feature of the settlements. From the bones of the cattle found in the settlements, and from the terracotta figurines and rock bruisings, the cattle appear to be lightly built, long-horned, humped animals of the *Bos indicus* breed, but not related to the heavier breeds depicted in the Indus culture area. An important and intriguing feature of this cattle-keeping culture, particularly during its earlier phase, were the huge ash mounds found at a number of sites. They were large heaps of accumulated dung that were periodically burnt, perhaps for some ritual purpose. Though some archaeologists would like to assign the ash mounds to the subsequent Iron Age, the avail-able evidence supports only a Neolithic date. It is, however, under-stood that dung-burning continued on a small scale up to the early Iron Age. There appears to be some continuity from one age to another for the symbolic importance of burning the dung. The suggestion that these ash mounds were associated with metal-working is not taken seriously now. Some difference of opinion as to the nature of the associated habitations exists. Some scholars believe that ash mounds were usually associated with nearby habitations, while others argue that certain ash mounds were located far from settlements and marked pastoral transhumance.

Cultivation of food plants is considered one of the most important traits of the Neolithic age. Some detailed knowledge of their agriculture is now available. The pulses horse gram, green gram, and the bigger millet and the pearl millet were the staple crops of the Neolithic community. Several other crops are attested to in the later Neolithic age: ragi, common foxtail millets, black and red grams, and hyacinth beans. Most millets associated with this culture were earlier believed to have been of African origin. Recent studies dispute this contention and suggest that a few food crops might have been introduced from Africa during the later stages, perhaps due to maritime contacts with the African coast. Ongoing studies suggest that the earliest crops were indigenous to southern India and domesticated locally, and those crops appear to have played an important role in the early agricultural economy of peninsular India. Later, in the second and first millennia BCE, crops of Near Eastern and African origins were integrated into, and in many cases came to dominate, local agricultural systems.

Of late, there have been some serious studies attempting to link the archaeo-botanical evidence and the ancient agricultural vocabulary in proto-Indo-Aryan and proto-Dravidian languages as

reconstructed using methods of historical linguistics. This was done with a view to understanding the identity and movements of the linguistic communities of the Neolithic and Iron Age cultures of India in general and of peninsular India in particular. These studies more or less concur that the Dravidian language group are the authors or originators of the southern Neolithic culture. There is, however, no unanimity of opinion on the question of whether this Neolithic community is indigenous to peninsular India or migrated from elsewhere and if they came from outside, from where and when. Further studies in archaeology and linguistics are necessary.

BEGINNINGS OF AGRICULTURE: NEOLITHIC CULTURE IN KERALA

Neolithic or the New Stone Age in human history was far more than a mere Stone Age. Apart from the fact that the people of the age used well-shaped and well-ground tools of smooth surfaces, they invented agriculture, domesticated animals, manufactured earthen pots, fabricated cloths and made use of fire for cooking. In many respects of culture, the Neolithic Age stands out totally distinct and far advanced than the other stone ages. As the age that marked the transition from food gathering to food production, ushering in the first major socio-economic transformation, the Neolithic phase is often conceived as a revolution.

Agriculture and cattle-rearing triggered a fundamental change in the living of the people. The assurance of food promoted a larger population, which in turn boosted production. The Neolithic Age marked the beginning of various handicrafts too. Earthenware of different sizes and shapes have been found among the relics of the Neolithic culture elsewhere. Clothes and ornaments made by people of this age have been found. A certain type of specialisation of arts and crafts might have existed across the scattered settlements and the people might have exchanged their products. As in the case of the Microlithic people, Neolithic people also had beliefs and rituals, but they were predominantly of the fertility cult. Fairs and festivals developed during this age, thereby bringing together these collectives around cult spots and facilitating exchanges.

Neolithic axes were made with gneiss or granite rock that was locally available. Polished celts (hand axes) are the tool types typical of the Neolithic Age across the world, though the raw material differed from place to place. The tool technology evident in stone axes, flakes, blades and beads seems to have spread all over the world during this age. Stone axes were made through flaking, pecking, grinding and polishing. Finished axes were symmetrical in shape and their working edge was well polished. Some of the axes bear marks of use at the convex working edge on the right angle. Antiquities of the world Neolithic culture date back to about 12,000 years before the present and those from the earliest Indian site date to about 5,000 years before present. In south India, the oldest known Neolithic site is dated back to 4,000 years before rock-art depictions, and some stratified indications here and there, we have no archaeological evidence of from roughly datable strata or layers on the walls of an excavated habitation sites of the Neolithic culture in Kerala. There is no reliable dating of these remains either. Comparing them with remains from the neighbouring regions for relative dating seldom helps because their features vary from site to site. However, it may be presumed that Neolithic people might have reached Kerala from Tamil Nadu and Karnataka sooner or later. This presumption of migration is drawing from the absence of habitation sites and the discovery of Neolithic tools at a few places, obviously brought by migrants from the settlements in the neighbouring states. The archaeologist Philip Lake was the first to discover Neolithic axes in Kerala in 1891. He discovered them at the foot of the Kannyakad hills. Later, William Logan (1887) and Fred Fawcett (1901) had collected

a few quartz flakes, a fragment of a Neolithic celt and a couple of beads from the Wayanad area. Polished stone axes and beads that are typical of the Neolithic phase were then reported from Kalpetta and Pulppalli in the Wayanad district and the riverbeds of the Periyar in Alwaye. Hand axes were collected as stray finds from Thamarassery, Puthady, Parambikkulam and Kundurmedu. Though very limited in number, the stray finds of the Neolithic hand axe in Kerala shows distribution across all the three physiographical zones, viz., the lowlands, midlands and highlands. Most of these pieces of evidence come from the surface, seemingly as objects carried away by people from other places, precluding stratified contexts and the possibility of fixing provenance. Also, it is probable that some of the artistic representations on walls of the rock-shelters of Eatakal, Ankode and Marayur are a product of the Neolithic culture.

Neolithic artefacts discovered in Kerala show that the culture had obviously got diffused from the sites of the neighbouring states and evolved here almost parallel to the development in other regions of south India. Despite differences in the raw material, Neolithic hand axes found in Kerala represent the typology typical of the culture elsewhere. In spite of the dominance of hand axe among the tool types discovered, certain features peculiar to the ecosystems-hilly forests, water-saturated swamps, river valleys, water bodies, high altitude grasslands, etc.-had evolved here. Nevertheless, there is no way to reconstruct the culture exactly, as it evolved out of the material processes of human adaptation to the ecological conditions of Kerala.

The principal means of subsistence all over the world during the Neolithic Age was agriculture or stock-rearing as determined by the ecosystems. In south India, there is Neolithic evidence for both pastoral as well as agricultural means of subsistence under appropriate ecosystems. In Kerala, the Neolithic means of subsistence was presumably a combination of gathering and primitive cultivation, supplemented by hunting/fishing. However, stock-rearing was never a major subsistence strategy even in highland Kerala that has no dearth of pastures. Extensive pastures in Kerala are confined to the high-altitude mountain ranges of Idukki district, and this accounts for indications of pastoral base in the Neolithic paintings at Marayur that depict stock-rearing corroborated by the stratified deposits around the time of the paintings.

Total absence of habitation sites and excavated material keeps us ignorant of the ethnic, organisational, institutional, customary and ritual aspects of the Neolithic material culture. However, in the light of existing knowledge about the culture in neighbouring regions, we assume that the earliest kin-based village settlements (ūr) in Kerala date back to the Neolithic phase. Probably the earliest Proto-Dravidian-speaking communities of Kerala must have lived in these villages. It is clear that the Neolithic people here as elsewhere had cultivated the slopes of the hills after burning the natural vegetation. This assumption is supported by the wild rice found in mid-forests and generic place names that mean 'rice fields'.

It appears that Neolithic culture was not widespread in Kerala, and that subsistence included a significant reliance on small animals such as the grey hare and tortise. The waterlogged and marshy landscape along midland like fishes and turtles, must have kept many Late Stone Age people confined to the Microlithic subsistence culture until the porcupine, monitors and crocodiles, and several aquatic creatures arrival of the iron-smelting people. In various places in Palakkad, Kozhikode, Wayanad, Thrissur and eastern parts of Ernakulam this seems to have

been the sequence of cultural transformation. It appears that the Neolithic phase of Kerala was very brief as it soon got exposed to the use of iron.

Conclusion

The archaeological evidence from South India and Kerala shows that human society evolved through a long and regionally varied process from the Paleolithic to the Neolithic period. During the Paleolithic and Mesolithic phases, communities in South India and Kerala depended largely on hunting, gathering, fishing, and the use of stone tools adapted to local environments such as river valleys, forests, rock shelters, and hill slopes. While South India presents clear and well-dated sequences of cultural development, especially in regions like the Krishna–Tungabhadra basin, Kerala's record remains fragmentary, dominated by surface finds of tools rather than stratified habitation sites. The Neolithic period in South India marks a major transformation with the introduction of agriculture, cattle-rearing, permanent settlements, polished stone tools, pottery, and ritual practices, reflecting a settled way of life and complex social organisation. In contrast, Kerala shows only limited and indirect evidence of Neolithic culture, suggesting that Neolithic traits diffused into the region from neighbouring areas rather than developing independently. The ecological conditions of Kerala—its dense forests, marshy lowlands, and rich aquatic resources—may have delayed or restricted the widespread adoption of agriculture, allowing microlithic and hunting-gathering traditions to persist longer. Overall, the prehistoric sequence of South India and Kerala highlights both continuity and diversity, demonstrating how human cultural development was shaped not only by technological innovation but also by regional environments and patterns of cultural interaction.

Model Questions

Maximum Time: 2 Hours Maximum Mark: 70

Section A

Answer all questions. Each question carries 3 marks.

(Ceiling 24 Marks)

1. Define the term “Palaeolithic.”
2. What is meant by the Pleistocene epoch?
3. What are hand-axes?
4. Define the Acheulian culture.
5. What is the Soan culture?
6. What are microliths?
7. What is meant by “hunter-gatherer society”?
8. Define the term “Mesolithic culture.”
9. What is meant by “rock shelter”?
10. Which are the major rock-art sites in Kerala.

Section B

Answer all questions. Each question carries 6 marks.

(Ceiling 36 Marks)

1. Explain the major characteristics of the Palaeolithic culture.
2. Analyse the relationship between environment and human life during the Palaeolithic period.
3. Discuss the regional distribution of Paleolithic tools in Keralam.
4. Examine the significance of rock art in understanding prehistoric Kerala.
5. Discuss the evidence for Upper Palaeolithic culture in Kerala.
6. Write a note on the evidence of Neolithic culture in Kerala.
7. Explain the importance of agriculture in the Neolithic Revolution.
8. Discuss the continuity and coexistence of Stone Age cultures in Kerala.

Section C

Answer any one question, carrying 10 marks.

(1 X10 = 10 Marks)

1. Examine the evidence for Paleolithic and Mesolithic cultures in Keralam.
2. Analyse the role of environmental conditions in shaping prehistoric cultures in Keralam.
3. Discuss the transition from hunting-gathering to food production during the Neolithic Age.
4. Evaluate the nature and extent of Neolithic culture in Keralam.
5. Discuss the importance of rock art, stone tools, and archaeological evidence in reconstructing the prehistoric past of Keralam.